Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2025

Name of the Course: B. Tech (Civil Engineering) Semester: III
Name of the Paper: Emerging Technologies in Construction Practices
Paper Code: TCE339
Time: 3 Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	**(20marks)**	
(a)	What is the role of the circular economy in the built environment? Explain how construction materials and waste can be managed within a circular economic model.	
(b)	Differentiate between green buildings, net-zero buildings, and climate positive buildings. What design features help achieve climate positivity?	**CO1**
(c)	What are the main components of Construction 4.0? Discuss how robotics, 3D printing, and digital twins can enhance efficiency, quality, and sustainability in the construction sector.	
Q2	**(20 marks)**	
(a)	What is deconstruction design, and how does it support circular economy principles in the construction industry? Discuss strategies for designing buildings with end-of-life reuse and recyclability in mind.	
(b)	Explain how landscape design and vegetation can enhance the sustainability and microclimate of a building site. Include examples of design interventions.	**CO2**
(c)	Discuss the role of building-integrated renewable energy systems (such as solar PV or solar thermal) in achieving net-zero or low-energy buildings.	
Q3	**(20 marks)**	
(a)	Define the concept of a Digital Building Life Cycle. How do Augmented Digital Design, Connected Construction, and Smart Operations contribute to achieving a fully digital and sustainable built environment?	**CO3**
(b)	Explain the role of Nanotechnology in improving the performance, sustainability, and durability of construction materials. Give suitable examples.	
(c)	Discuss how Artificial Intelligence (AI), Machine Learning (ML), and Big Data Analytics are used for predictive maintenance, resource optimization, and risk management in construction projects.	
Q4	**(20 marks)**	
(a)	Discuss the role of BIM across the construction value chain. How does BIM integrate design, construction, and facility management processes?	**CO4**
(b)	Explain how BIM tools facilitate clash detection and coordination among different building systems (structural, architectural, MEP).	
(c)	What are collaborative practices and standardization protocols (such as ISO 19650) that ensure effective BIM-based project delivery? Discuss their importance in digital construction workflows.	
Q5	**(20 marks)**	
(a)	Discuss the applications of Drones, Wearable Devices, and AR/VR/MR technologies in improving safety and productivity on construction sites.	**CO5**
(b)	Compare Prefabrication, Robotic Fabrication, and Additive Manufacturing in modern building construction techniques.	
(c)	What are High-Performance Insulators? Explain their types, properties, and significance in energy-efficient buildings.	